BJT operation and DC bias I

Answers and worked solutions · Day 06 · Thu 29 Oct · pack E4

5. Practice

From your course files

WS4 Q6 (course question).

- **6.1 (A).** $i_C = I_S e^{v_{BE}/V_T}$: the base–emitter voltage controls the collector current. A PNP has p-type emitter and collector, and a BJT has two junctions, not three.
- **6.2 (C).** In cut-off both junctions are reverse biased. Option C names the emitter-base junction. (A is the same junction written as “base-emitter”, but the official answer is C.)
- **6.3 (D, official).** The official solution reads D as “electrons flow from emitter to base”, which is the correct physics: the forward-biased B–E junction injects electrons from the n-type emitter into the base. On the question sheet, options B and D are both printed as “holes flow from emitter to base”, so D as printed is wrong for an NPN. Answer with the physics: electrons flow from emitter into the base (and only *indirectly* on to the collector, so C is wrong).

WS4 Q7 (course question).

$\beta = \dfrac{\alpha}{1-\alpha}$: $\dfrac{0.5}{0.5} =$ **1**, $\dfrac{0.8}{0.2} =$ **4**, $\dfrac{0.98}{0.02} =$ **49**, $\dfrac{0.999}{0.001} =$ **999**. β grows very fast as α approaches 1.

WS4 Q8 (course question).

1. $I_B = I_C/\beta = 4/50 =$ **0.08 mA**, $I_E = I_C + I_B =$ **4.08 mA**.
2. $\alpha = \beta/(1+\beta) = 50/51 = 0.98$, $I_C = \alpha I_E =$ **3.92 mA**, $I_B = I_E - I_C =$ **0.08 mA**.
3. $I_C = \beta I_B = 50 \times 0.1 =$ **5 mA**, $I_E = I_C + I_B =$ **5.1 mA**.

WS4 Q9 (course question).

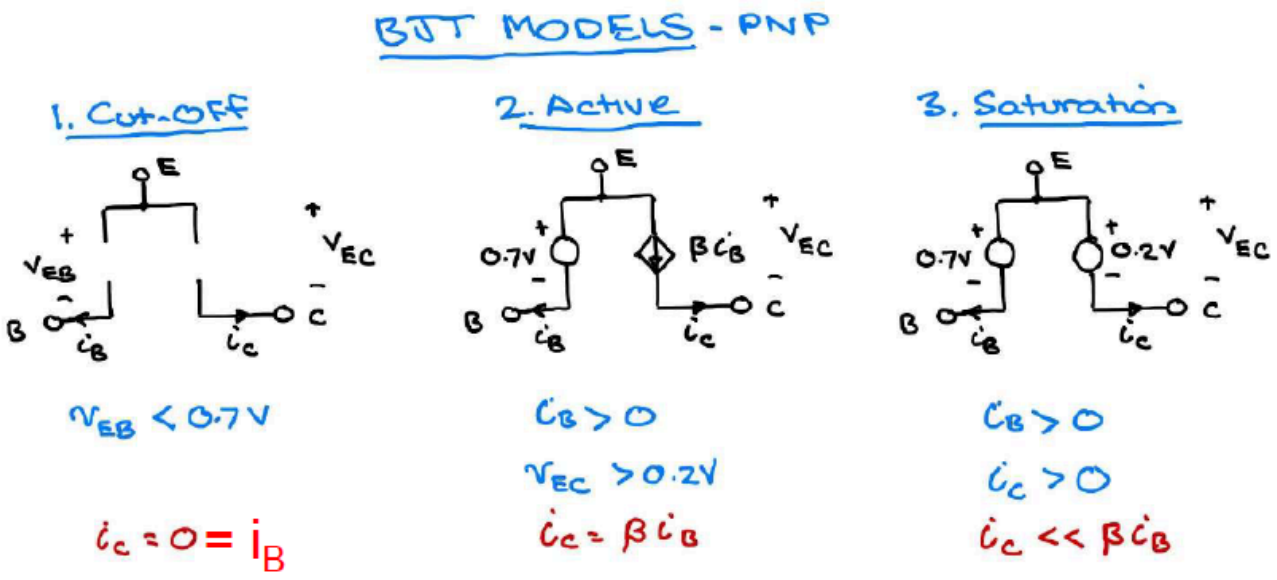
1. $\beta = I_C/I_B = 8/0.1 =$ **80**, $\alpha = \beta/(1+\beta) =$ **0.988**, $I_E = I_C + I_B =$ **8.1 mA**.
2. Part 2 is Worked example 1 above: $I_C =$ **4.9 mA**, α = **0.98**, β = **49**.
3. $I_B = I_E - I_C =$ **0.1 mA**, $\alpha = 10/10.1 =$ **0.9901**, $\beta = 10/0.1 =$ **100**.

WS4 Q10 (course question).

Worked in Worked example 1 above (the table): **cut-off, active, saturation, active.** Compare I_C with βI_B: equal means active, smaller means saturation, and $I_B = 0$ means cut-off.

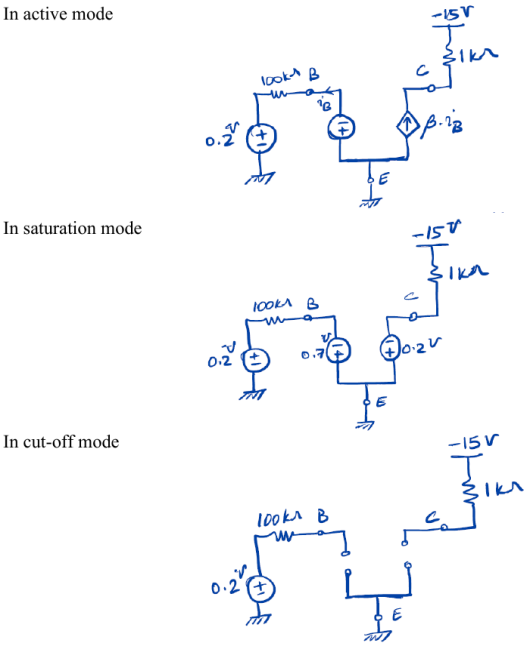
WS4 Q11 (course question).

Use the PNP models from the sheet (Figure 5): active is $V_{EB} = 0.7$ V between E and B plus βi_B from E to C; saturation is 0.7 V (E–B) plus 0.2 V (E–C); cut-off is open between all terminals.



WS4 Figure 5: PNP large-signal models (from the workshop sheet)

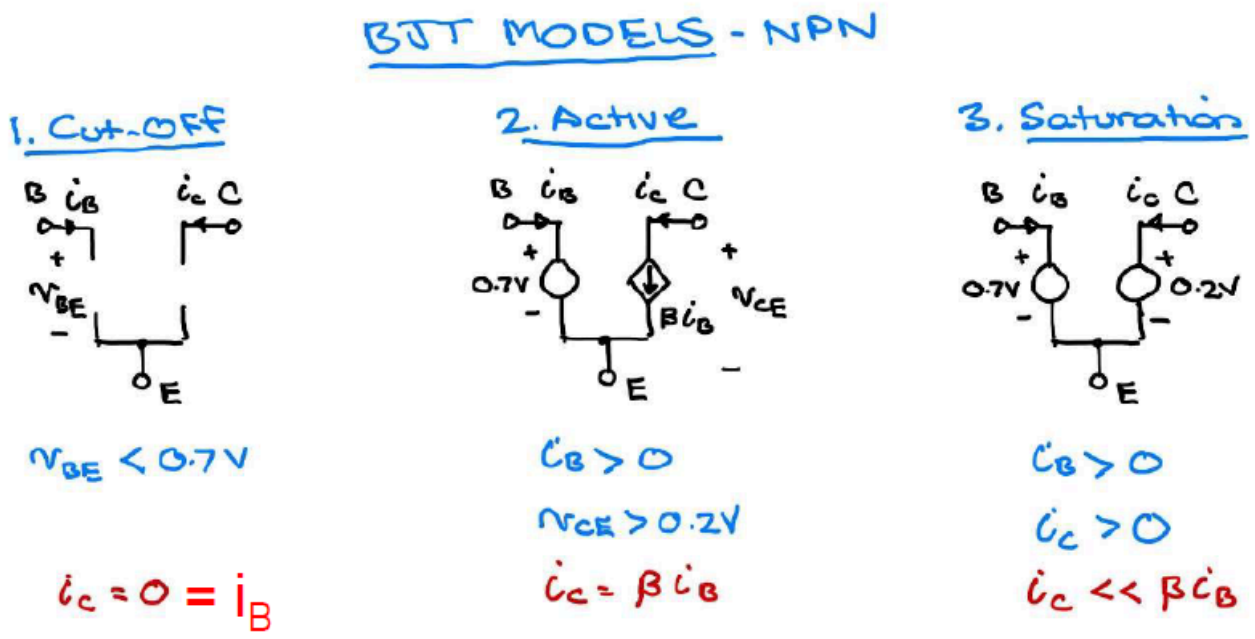
Applied to this circuit (emitter to ground, base through 100 kΩ to 0.2 V, collector through 1 kΩ to −15 V):



WS4 Q11 official redraws in the three modes

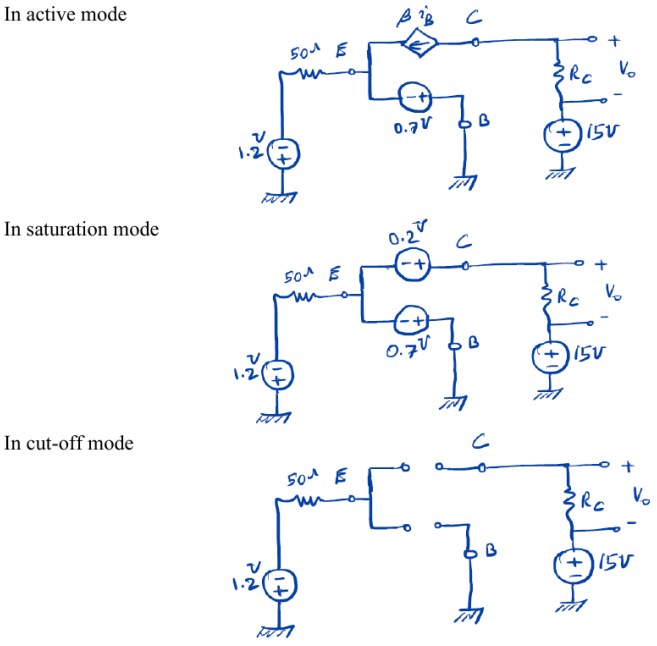
WS4 Q12 (course question).

Use the NPN models from the sheet (Figure 4): active is $V_{BE} = 0.7$ V between B and E plus βi_B from C to E; saturation is 0.7 V (B–E) plus 0.2 V (C–E); cut-off is open.



WS4 Figure 4: NPN large-signal models (from the workshop sheet)

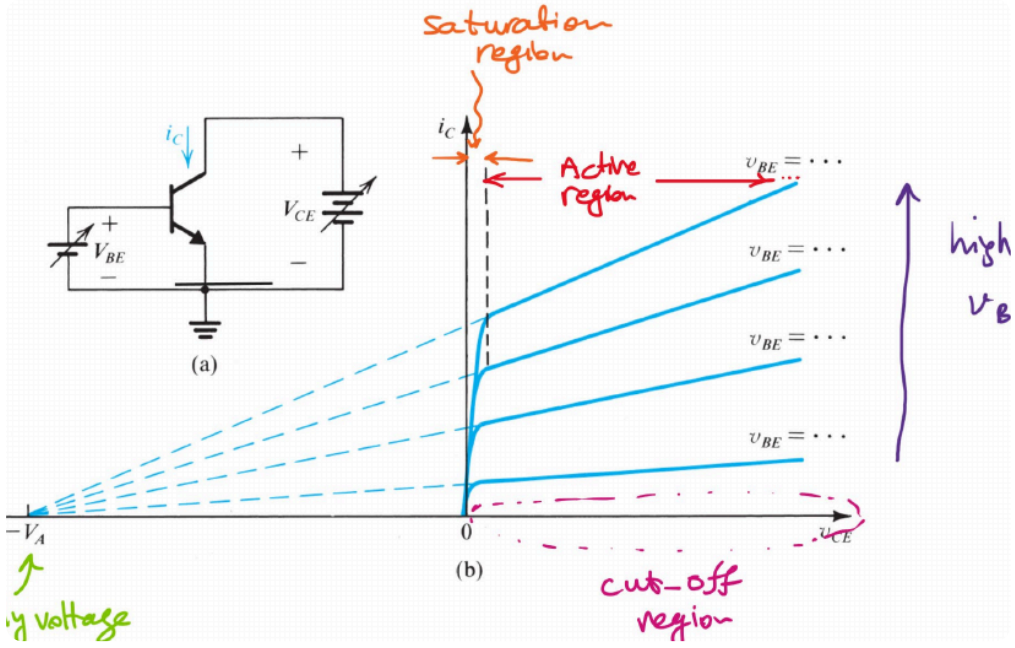
Applied to this common-base circuit (emitter through 50 Ω to 1.2 V, base grounded, collector through R_C to 15 V):



WS4 Q12 official redraws in the three modes

WS5 Q1 (course question).

- **Saturation:** the steep part near $v_{CE} \approx 0$, left of the dashed line. **Active:** the sloped, nearly flat part to the right. **Cut-off:** along the v_{CE} axis, $i_C \approx 0$. **Early voltage:** the point $-V_A$ where the extended active-region lines meet on the v_{CE} axis.
- The non-zero slope means i_C still rises with v_{CE} at fixed v_{BE} (the Early effect, modelled by a resistor r_o in the model).
- Higher curves need higher v_{BE}: v_{BE} increases as i_C increases.



WS5 Q1 official annotated answer

(The question says $i_c - v_{CB}$, but the axes on the figure are i_C against v_{CE}.)

WS5 Q3 (course question).

Hint: no α or β is given, so get I_B from the KVL round the loop that contains the 50 kΩ.

1. $I_E = \dfrac{3 - V_E}{5} =$ **0.4 mA**. Loop +3 V → 5 kΩ → E → B → 50 kΩ → ground: $-3 + 5I_E + 0.7 + 50I_B = 0$, so $I_B = \dfrac{3 - 0.7 - 2}{50} =$ **6 µA**. $I_C = I_E - I_B =$ **0.394 mA**.
2. $V_B = 50I_B =$ **0.3 V** (check: $V_E - V_B = 0.7$ V). $V_C = -3 + 5I_C =$ **−1.03 V**.
3. $\alpha = I_C/I_E =$ **0.985**, $\beta = \alpha/(1-\alpha) =$ **65.7**.

WS5 Q4 (course question).

$R_C = \dfrac{1.5 - V_C}{I_C} = \dfrac{1.5 - 0.5}{0.2} =$ **5 kΩ**. The base is grounded, so $V_E = -0.7$ V.
$\alpha = 100/101 = 0.9901$ and $I_E = I_C/\alpha = 0.202$ mA.

$$R_E = \frac{V_E - (-1.5)}{I_E} = \frac{0.8}{0.202} = \mathbf{3.96\ k\Omega}$$

WS5 Q5 (course question).

1. Emitter loop: $-2.7 + 0.2I_E + 0.7 = 0$, so $I_E = \mathbf{10\ mA}$. $I_C = \alpha I_E = \mathbf{9\ mA}$, $I_B = I_E - I_C = \mathbf{1\ mA}$.
2. Outer loop: $-2.7 + 0.2I_E + V_{EC} + R_C I_C - 20 = 0$, so $R_C = \dfrac{20.7 - V_{EC}}{9}$ (kΩ, mA). The smallest R_C goes with the largest V_{EC}: $R_{C,min} = \dfrac{20.7 - 20}{9} = \mathbf{77.8\ \Omega}$.

WS5 Q6 (course question).

1. Emitter loop: $+0.7 + 0.05I_E - 1.2 = 0$, so $I_E = 10$ mA and $I_C = 0.9 \times 10 = 9$ mA. $R_C = |V_o|/I_C = 4.5/9 = \mathbf{0.5\ k\Omega}$.
2. Outer loop: $-15 + R_C I_C + V_{CE} + 0.05I_E - 1.2 = 0$, so $V_{CE} = 15 - 4.5 - 0.5 + 1.2 = \mathbf{11.2\ V}$.

WS5 Q7 (course question).

Worked in full in Worked example 3 above. The $\beta R_E \geq 10R_2$ test fails (54.4 k < 91 k), so use the exact Thévenin method: (a) **21.4 µA**, (b) **1.71 mA**, (c) **8.15 V**, (d) **9.33 V**, (e) **1.18 V**, (f) **1.88 V**.

WS5 Q8 (course question).

Test: $\beta R_E = 120 \times 1 = 120$ kΩ $\geq 10R_2 = 82$ kΩ, so the approximate approach is allowed.

- e. $V_B = 18 \times \dfrac{8.2}{39 + 8.2} = \mathbf{3.13\ V}$
- d. $V_E = V_B - 0.7 = \mathbf{2.43\ V}$
- a. $I_C \approx I_E = V_E/R_E = \mathbf{2.43\ mA}$
- b. $V_{CE} = 18 - 2.43(3.3 + 1) = \mathbf{7.55\ V}$ (7.56 V unrounded; the official 7.55 V rounds I_C first)
- c. $I_B = I_C/\beta = \mathbf{20.2\ \mu A}$ (official 20.25 µA)

WS5 Q14 (course question).

Thévenin first (this is the carry-on in pack E1, Example 1): $V_{Th} = 20 \times \dfrac{10}{100} = 2$ V, $R_{Th} = 90 \parallel 10 = 9$ kΩ.

Assume active. Base loop: $-2 + 9I_B + 0.7 + 1 \cdot (\beta+1)I_B = 0$, so $I_B = \dfrac{1.3}{9+51} =$ **21.7 µA** and $I_C = 50 I_B =$ **1.08 mA**. $I_E = I_C + I_B = 1.107$ mA.

Collector loop: $V_{CE} = 20 - 5I_C - 1 \cdot I_E = 20 - 5.42 - 1.11 =$ **13.5 V**.

> **CHECK:** $I_B > 0$ and $V_{CE} = 13.5 > 0.2$ V, so the transistor is **active**.

PT Q6 (practice test, 6 marks).

This is emitter-stabilised bias: one base resistor from V_{CC} plus an emitter resistor.

Load line. Output loop: $V_{CE} = V_{CC} - I_C(R_C + R_E)$ with $R_C + R_E = 0.47 + 2.2 = 2.67$ kΩ.

- $I_C = 0$ gives $V_{CE} = 20$ V.
- $V_{CE} = 0$ gives $I_{C,sat} = 20/2.67\ \text{k} = 7.49$ mA.

Join (20 V, 0) to (0, 7.49 mA) with a ruler.

Base current (this decides which curve to use):

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{20 - 0.7}{270 + 126(2.2)} = \frac{19.3}{547.2\ \text{k}} = 35.3\ \mu\text{A}$$

(a) Q-point. 35.3 µA lies between the 30 µA and 40 µA curves, about half-way. Near $V_{CE} \approx 8$ V the printed curves sit at about 3.9 mA (30 µA) and 5.05 mA (40 µA). Interpolating gives $I_C \approx 3.9 + 0.53(1.15) \approx 4.5$ mA. The load line passes through 4.5 mA at $V_{CE} = 20 - 4.5(2.67) = 8.0$ V. So read:

$$I_{CQ} \approx 4.5\ \text{mA}, \qquad V_{CEQ} \approx 8.0\ \text{V}$$

Analytic cross-check with β = 125: $I_{CQ} = 125 \times 35.3\ \mu\text{A} = 4.41$ mA, $I_E = 4.44$ mA and $V_{CEQ} = 20 - 4.41(0.47) - 4.44(2.2) = 8.15$ V. These agree with the graph to within reading accuracy. Quote the graph values, because the question asks for the Q-point on the characteristics.

(b) β at the Q-point.

$$\beta_{dc} = \frac{I_{CQ}}{I_{BQ}} \approx \frac{4.5\ \text{mA}}{35.3\ \mu\text{A}} \approx 127$$

This is consistent with the printed β = 125. Write the two numbers you read and then the division.

Cross-check with your Test 1 Reference. The reference has the same load line, I_{BQ} and analytic Q-point (4.41 mA, 8.15 V). For the graph it reads I_{CQ} as roughly 4.2–4.4 mA and gets β ≈ 120. I measured the printed curves from the PDF image: the 30 µA and 40 µA curves sit

at about 3.86 mA and 5.05 mA near 8 V, which puts the crossing nearer 4.5 mA and gives β ≈ 127. Either answer, in the range of about 120–130, should earn the marks if you show the readings. The spacing of the curves, $\Delta I_C/\Delta I_B \approx 1.19\ \text{mA}/10\ \mu\text{A} \approx 119$, is β_{ac}, not what part (b) asks for. The Practice Test pack (EPT) reads the same graph slightly differently (about 4.6 mA, 7.6 V, β ≈ 130). Treat **4.5–4.6 mA, 7.6–8.0 V, β ≈ 127–130** as the graph answer; any reading in that band, with the readings shown, is fine.

PT Q7 (practice test, 10 marks).

(a) Thévenin equivalent seen from the base.

Resistance. Short both sources to ground. R_1 and R_2 are then in parallel:

$$R_{Th} = R_1 \parallel R_2 = \frac{15 \times 30}{45} = 10\ \text{k}\Omega$$

Voltage. With the base open there are two sources, so use superposition:

$$V_{Th} = 9 \cdot \frac{30}{15+30} + (-12) \cdot \frac{15}{15+30} = 6 - 4 = 2\ \text{V}$$

Check by inspection: the base node sits $30/45$ of the way from −12 V up to +9 V, so $-12 + 21 \times \frac{2}{3} = 2$ V. ✓

$V_{Th} = 2$ **V and** $R_{Th} = 10$ **kΩ.**

(b) Redraw. Draw a 2 V source in series with 10 kΩ feeding the base. Between base and emitter, draw a 0.7 V source (+ at the base). Between collector and emitter, draw a current source $\beta I_B = 50 I_B$ pointing from collector to emitter. Connect the emitter to ground (the common node). Connect the collector to 12 V through 1.5 kΩ, and label V_C at the collector and I_B and I_C with arrows. For the marks, label every element value and the current directions.

Practice Test Q7(b) redrawn: Thévenin base source and the active-mode DC model.

(c) Currents and collector voltage. The emitter is grounded, so there is no $(\beta+1)R_E$ term:

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th}} = \frac{2 - 0.7}{10\ \text{k}} = 130\ \mu\text{A}, \qquad I_C = \beta I_B = 50 \times 130\ \mu\text{A} = \mathbf{6.5}\ \text{mA}$$

$$V_C = V_{CC} - I_C R_C = 12 - 6.5(1.5) = \mathbf{2.25}\ \text{V}$$

Check: $V_{CE} = V_C = 2.25 > 0.2$ V and $I_B > 0$, so the active assumption holds.

This matches your Test 1 Reference exactly. The method is the same as WS5 Q14. The only new step is the superposition for V_{Th}.

New questions

New question 1

a. $\alpha = \beta/(\beta+1) = 99/100 = \mathbf{0.99}$.

b. $I_{CEO} = I_{CBO}/(1-\alpha) = 10\ \text{nA}/0.01 = \mathbf{1}\ \boldsymbol{\mu}\mathbf{A}$, which equals $(\beta+1)I_{CBO}$.

c. $I_C = \beta I_B + I_{CEO} = 99(20\ \mu\text{A}) + 1\ \mu\text{A} = 1.980 + 0.001 = \mathbf{1.981}\ \text{mA}$.
$I_E = I_C + I_B = \mathbf{2.001}\ \text{mA}$. The leakage is 0.05 % of I_C, so it is negligible at room temperature. That is why we normally write $I_C = \beta I_B$. It becomes important at high temperature, because I_{CBO} roughly doubles every 10 °C.

New question 2

a. $I_B = (15 - 0.7)/470\ \text{k} = \mathbf{30.4}\ \boldsymbol{\mu}\mathbf{A}$, $I_C = 100 \times 30.4\ \mu\text{A} = \mathbf{3.04}\ \text{mA}$, and $V_{CE} = 15 - 3.04(3.3) = \mathbf{4.96}\ \text{V}$.

b. The load-line intercepts are $V_{CE} = 15$ V (cut-off) and $I_{C,sat} = 15/3.3\ \text{k} = \mathbf{4.55}\ \text{mA}$. $V_B = 0.7$ V and $V_C = 4.96$ V, so $V_{BC} = 0.7 - 4.96 = -4.26$ V. The CBJ is reverse-biased and the EBJ is forward-biased, so the transistor is **active**. ✓ The Q-point sits at 3.04/4.55 = 67 % of the way up the load line, towards saturation.

New question 3

a. Base loop: $I_B = \dfrac{18 - 0.7}{510 + 101(1.5)} = \dfrac{17.3}{661.5\ \text{k}} = \mathbf{26.2}\ \boldsymbol{\mu}\mathbf{A}$. $I_C = \beta I_B = \mathbf{2.62}\ \text{mA}$ and $I_E = 101 I_B = 2.64$ mA.
$V_{CE} = 18 - 2.615(2.4) - 2.641(1.5) = 18 - 6.28 - 3.96 = \mathbf{7.76}\ \text{V}$.

b. $V_E = I_E R_E = \mathbf{3.96}$ V, $V_B = V_E + 0.7 = \mathbf{4.66}$ V, $V_C = 18 - 6.28 = \mathbf{11.72}$ V, and $V_{BC} = 4.66 - 11.72 = \mathbf{-7.06}$ V. The CBJ is reverse-biased, so the transistor is active. ✓

c. $I_{C,sat} = V_{CC}/(R_C + R_E) = 18/3.9\ \text{k} = \mathbf{4.62}\ \text{mA}$.

Note that the reflected emitter resistance $(\beta+1)R_E = 151.5$ kΩ is a significant part of the 661.5 kΩ in the base loop.

New question 4

a. $\beta R_E = 150 \times 1.2 = 180$ kΩ and $10R_2 = 100$ kΩ. Since 180 ≥ 100, the approximation is **valid**.

b. $V_B = 15 \times 10/57 = 2.632$ V and $V_E = 1.932$ V.
$I_C \approx I_E = 1.932/1.2 = \mathbf{1.61}$ mA. $V_{CE} \approx 15 - 1.61(2.7 + 1.2) = \mathbf{8.72}$ V.
$I_B = 1.61/150 = \mathbf{10.7}\ \boldsymbol{\mu}\mathbf{A}$.

c. $R_{Th} = 47 \parallel 10 = 8.25$ kΩ and $E_{Th} = 2.632$ V.
$I_B = \dfrac{2.632 - 0.7}{8.25 + 151(1.2)} = \dfrac{1.932}{189.5\ \text{k}} = \mathbf{10.2}\ \boldsymbol{\mu}\mathbf{A}$. $I_C = 150 I_B = \mathbf{1.53}$ mA and $I_E = 1.54$ mA. $V_{CE} = 15 - 1.529(2.7) - 1.539(1.2) = \mathbf{9.02}$ V. The approximation over-estimates I_C by $(1.610 - 1.529)/1.529 = \mathbf{5.2}\ \%$. Even when the test passes, expect a few percent of error.

New question 5

a. $\beta R_E = 60 \times 0.56 = 33.6$ kΩ and $10 R_2 = 120$ kΩ. Since 33.6 < 120, the approximation is **not valid**.

b. Approximate: $V_B = 12 \times 12/45 = 3.2$ V, $V_E = 2.5$ V, and $I_C \approx 2.5/0.56 = 4.46$ mA. Then $V_{CE} = 12 - 4.46(2.76) = -0.32$ V. A negative V_{CE} is impossible in active mode. The student would wrongly conclude that the transistor is saturated. Also, 4.46 mA exceeds $I_{C,sat}$ (part d), which no active transistor can do. The base current is not negligible here, so the divider is heavily loaded and V_B is really much lower than 3.2 V.

c. Exact: $R_{Th} = 33 \parallel 12 = 8.8$ kΩ and $E_{Th} = 3.2$ V.

$$I_B = \frac{3.2 - 0.7}{8.8 + 61(0.56)} = \frac{2.5}{42.96\ \text{k}} = \mathbf{58.2}\ \boldsymbol{\mu}\mathbf{A}$$

$I_C = 60 I_B = \mathbf{3.49}$ mA and $I_E = 61 I_B = 3.55$ mA.
$V_{CE} = 12 - 3.492(2.2) - 3.550(0.56) = \mathbf{2.33}$ V. $V_E = \mathbf{1.99}$ V, $V_B = \mathbf{2.69}$ V (compare 3.2 V unloaded), and $V_C = 12 - 7.68 = \mathbf{4.32}$ V. Mode: $V_{CE} = 2.33 > 0.2$ V and $V_{BC} = 2.69 - 4.32 = -1.63$ V, so the transistor is **active**. The approximation was 28 % high on I_C.

d. $I_{C,sat} = 12/(2.2 + 0.56) = \mathbf{4.35}$ mA. The Q-point is at $I_C/I_{C,sat} = 3.49/4.35 = 80\ \%$ of the way up the load line, with $V_{CE} = 2.33$ V out of 12 V. It is well towards saturation, which is a poor bias point for symmetrical signal swing.

New question 6

a. $V_{CC} = 24$ V (the cut-off intercept). $R_C + R_E = V_{CC}/I_{C,sat} = 24/8$ mA $= \mathbf{3\ k\Omega}$, so $R_E = 3 - 2.4 = \mathbf{0.6\ k\Omega}$.

b. $V_{CEQ} = 24 - 4.2(3) = \mathbf{11.4}$ V and $\beta_{dc} = 4.2$ mA$/30\ \mu$A $= \mathbf{140}$.

c. $I_E = (\beta + 1) I_B = 141 \times 30\ \mu\text{A} = 4.23$ mA, so $V_E = 4.23(0.6) = 2.54$ V. Then

$$R_B = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{24 - 0.7 - 2.54}{30\ \mu\text{A}} = \mathbf{692\ k\Omega}$$

The nearest E12 value is **680 kΩ**. Then $I_B = \dfrac{23.3}{680 + 141(0.6)} = \dfrac{23.3}{764.6\ \text{k}} = \mathbf{30.5\ \mu A}$, which is 1.6 % above the target. That is fine.

6. Quick recall check

Question

1. $\beta = 0.995/0.005 = 199$ and $\alpha = 150/151 = 0.9934$.
2. $V_{BE} = 0.7$ V, so the EBJ is forward-biased. $V_{CE} = 0.2$ V, which means $V_{BC} = +0.5$ V and the CBJ is also forward-biased. Both forward means **saturation**.
3. $I_B = \dfrac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$.
4. $\beta R_E \geq 10 R_2$. The resistance looking into the base (about βR_E) is then large compared with R_2, so the base current does not load the divider and $V_B \approx R_2 V_{CC}/(R_1 + R_2)$.
5. $(V_{CE}, I_C) = (V_{CC},\ 0)$ and $(0,\ V_{CC}/(R_C + R_E))$.